How to use R with your Data storage in Google Drive

Unlock your data's potential with R and Google Drive

1. STORE DATA IN GOOGLE DRIVE → 2. CONNECT & SETUP IN COLAB → 3. RUN R CODE IN NOTEBOOK → 4. ANALYZE YOUR DATA → 5. INSIGHTS & DECISIONS

1. STORE YOUR DATA IN GOOGLE DRIVE

- My Drive
- Colab Notebooks
- Data
- Reports

Folder contains the CSV file with prices of sowing in Peru (Jan 2020 – Jul 2021)

PERU.GOB.PE.csv

1. EXAMPLE: INSTALL & LOAD R PACKAGES

```
%%R
install.packages('reshape2')    library(reshape2)
install.packages('lubridate')   library(lubridate)
library(plyr)
library(tidyverse)
library(ggplot2)
library(data.table)
install.packages('DBI')         library(DBI)
install.packages('RMySQL')      library(RMySQL)
```

✓ You can install and load any R package just like in your local R environment!

2. NOTE: R MAGIC COMMAND NOT SUPPORTED

In the latest version in Colaboratory doesn't support R magic command (updated: December 19, 2022)

```
%%R
Error: Cell magic `%%R` not found.
```

3. SETUP R IN COLAB (USE A COMPATIBLE rpy2 VERSION)

```
!pip install -q rpy2==3.5.1     # install a working version
%load_ext rpy2.ipython          # enable R magic command
```

R magic commands:

%R runs a single line of R code

```
%R data <- c('banana')
```

%%R runs multiple lines of R code

```
%%R
install.packages('ggplot2')
library(ggplot2)
```

4. CONNECT GOOGLE DRIVE & SET WORKING DIRECTORY

```
from google.colab import drive
drive.mount('/content/drive')
```

```
path =
"/content/drive/My Drive/Colab Notebooks/"
%cd {path}
```

```
!ls
PERU.GOB.PE.csv
notebook.ipynb
analysis.R
data/
```

✓ Mount Google Drive

✓ Set file location & change directory

5. DATA ANALYTICS WITH R

① Read the CSV file

```
%%R
precios <- read.csv(file = 'PERU.GOB.PE.csv', sep=',')
```

```
head(precios,1)
 Producto   Jan.20   Feb.20   Mar.20   Apr.20   ...
  Arroz       2.48     2.46     2.40     2.35   ...
```

② Display all columns

```
%%R
names(precios)
```

```
output:
[1] "Producto"  "Jan.20"  "Feb.20"  "Mar.20"  ...
```

③ Lower case product names

```
%%R
precios$Producto <- tolower(precios$Producto)
head(precios)
```

```
output:
  producto   jan.20   feb.20   mar.20   apr.20   ...
1  arroz       2.48     2.46     2.40     2.35   ...
```

④ Melt by Producto & convert Fecha

```
%%R
library(reshape2)
precios_melt <- melt(precios, variable.name = "Fecha", value.name = "Precio")
precios_melt$Fecha = as.Date(parse_date_time(precios_melt$Fecha, "%b.%y"))
names(precios_melt) <- toupper(names(precios_melt))
str(precios_melt)
```

```
output: (structure)
'data.frame':  29582 obs. of  3  variables:
 $ PRODUCTO :  chr  "arroz" "arroz"  ....
 $ FECHA    :  Date, format: "2020-01-01" 4..
 $ PRECIO   :  num  2.48 2.46 2.4 2.35 ...
```

⑤ Clean the data

```
%%R
precios_melt <- na.omit(precios_melt)
precios_melt$PRECIO <- round(precios_melt$PRECIO, 2)
precios_melt <- precios_melt[precios_melt$PRECIO > 0,]
head(precios_melt)
```

```
output:
PRODUCTO    FECHA         PRECIO
 arroz      2020-01-01      2.48
 arroz      2020-02-01      2.46
 arroz      2020-03-01      2.40
 arroz      2020-04-01      2.35 ...
```

CONCLUSION Colaboratory allows you to run R scripts without installing anything on your machine, and with Google Drive you can access and analyze your data anywhere, anytime!

You can find the Colaboratory notebook here. Open in Colab